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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	404 Sleep Not Found
Selected Domain	DOMAIN 2: ALPHA RED
Selected SDG Track (1–17)	8: Decent Work and Economic Growth
Project / Solution Name	StockSync AI for Orbyte
Name (Leader)	Ling Xin Qi
Email & Phone (Leader)	xinqi052@gmail.com & 0188704593
University Affiliation(s)	Universiti Sains Malaysia
Number of Team Members	3

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Small and Medium Enterprises (SMEs) are an important part of local communities, but many still face challenges in managing inventory and procurement. Store managers often rely on manual stock-taking, visual estimates, and informal communication with suppliers, making it difficult to respond quickly to changes in customer demand.

This can lead to two major problems:

1. **Stockouts:** Popular products can run out unexpectedly when demand increases faster than expected. This results in missed sales, dissatisfied customers, and disruptions in the supply process.
2. **Overstock and Waste:** To avoid stockouts, managers may order more than necessary, especially for slow-moving products. This ties up limited working capital and increases storage costs and, for perishable goods, product waste.

Procurement can also be time-consuming. Managers may need to manually prepare purchase orders, calculate order quantities and costs based on different supplier requirements, and send separate messages or emails to suppliers. For small businesses with limited staff, these repetitive tasks can take up valuable time that could instead be spent on serving customers and improving the business.

Therefore, there is a need for a more efficient approach to inventory and procurement management that can help SMEs reduce stockouts, avoid unnecessary overstock, and reduce the administrative workload involved in reordering.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

StockSync AI modernizes retail procurement through an automated workflow that combines predictive inventory management with intelligent ordering.

1. **Adaptive Weekly Demand & Replenishment Intelligence:** Instead of relying on fixed reorder points, StockSync AI analyzes customer sales over a rolling 7-day period to estimate daily product usage. The system uses ceiling division to round fractional daily usage up to the nearest whole unit, avoiding fractional stock calculations. When an item's estimated stock coverage falls to 7 days or less, the system automatically calculates the recommended replenishment quantity based on 30 days of estimated demand and alerts the store manager. Items with no sales during the past 7 days have an estimated daily usage of 0 and are excluded from replenishment recommendations.
2. **Context-Aware Procurement & Communication Agent:** Once the store manager approves a replenishment recommendation, StockSync AI automatically generates a supplier-specific purchase order draft. It retrieves relevant supplier information, calculates the required quantities and total costs, includes supplier lead times, and prepares a professional order letter.

The system follows a **Human-in-the-Loop** approach, allowing managers to review, edit, approve, or reject AI-generated orders before sending them directly to suppliers through integrated Gmail. Managers confirm order records and update order statuses to *Sent*, *Pending Delivery*, or *Received* based on actual delivery progress.

StockSync AI applies technical skills developed through SDC workshops, including TypeScript/Node.js, modern component architectures, schema-validated browser data persistence, and cloud deployment pipelines to deliver a practical web application.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

StockSync AI is a web-based inventory management prototype that monitors stock levels, analyses sales data, and generates restocking recommendations.

- **Frontend:** React, TypeScript, Vite, Tailwind CSS, and Recharts.
- **Backend:** Node.js and Express provide APIs for inventory, supplier, sales, and import data.
- **Data Storage:** Local JSON files store the prototype datasets. The server must be restarted to load manual changes.
- **Data Validation:** Zod validates dataset structures.
- **AI Analysis:** Python and FastAPI provide an endpoint for inventory analysis and restocking recommendations.
- **Development Platform:** Replit was used to develop the application with AI-assisted prompting, with code pushed to Git for version control.
- **Database:** No database is currently used for the main dataset. Local JSON files serve as the data source.

Sponsor-provided platform: Replit was used for development.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We adopted a user-centred design approach focused on retail managers with limited IT experience. Through iterative feedback, we recognized that fully automated procurement could pose financial risks to small businesses. Therefore, StockSync AI uses a **Human-in-the-Loop** workflow that combines AI automation with managerial control.

1. **Simple Visual Cues:** Inventory information is presented as clear AI-suggested tasks with status indicators, helping managers identify urgent actions without feeling overwhelmed.

2. **Simplified Procurement:** Managers can review an inventory alert and generate a structured email draft in just two clicks, reducing repetitive procurement tasks and saving time.
3. **Built-in Safeguards:** Rejection options, editable order templates, and audit trails help prevent accidental orders and maintain accountability.

For deployment, StockSync AI uses a lightweight client application supported by an Express.js microservice. Its simple architecture reduces setup requirements and makes it suitable for small retailers with limited technical resources. The application can also be containerized using Docker to support future cloud deployment and further development.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target 8.3: Promote entrepreneurship and the growth of MSMEs. StockSync AI helps small retailers improve inventory management, streamline procurement, reduce operational inefficiencies, and use their limited resources more effectively. Target 8.2: Achieve higher levels of economic productivity. StockSync AI automates repetitive inventory and ordering tasks, allowing store managers to spend less time on administrative work and more time on customer service and business operations.

Metric Category	Your Response
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Research shows that doing inventory by hand causes frequent errors, waste, and financial losses (Wynn & Kuhn, 2021). In the retail sector, this manual work wastes 4 to 6 hours of a manager's week (DCSME, 2026) and directly leads to a 12% to 15% out-of-stock rate for fast-moving consumer goods (IHL Group).
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	1. **Administrative Labor Saved:** Hours of manual inventory management time saved per manager, per week. 2. **Stockout Reduction:** Percentage decrease in fast-moving consumer goods (FMCG) out-of-stock occurrences. 3. **Procurement Efficiency:** Purchase Order (PO) processing time (measured in minutes per order).
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Achieve a 75% reduction in manual administrative labor (saving 3 to 4.5 hours per manager, per week), a 25% reduction in FMCG stockout occurrences, and an 80% decrease in purchase order processing time (from ~25 minutes down to < 5 minutes per PO) through automated, velocity-based reorder triggers.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ☑ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



22nd October 2026

Appendix

